

M5 How to reinterpret the clue key ID BY ROWS

So the instruction is: use jumps of 9 and 4 BY ROWS. We will name this algorithm combining 2 jumps, the TFJ algorithm. We have 6 possible jump patterns: IDID, DIDI, IIDD, DDII, IDDI and DIID. While the TFJ algorithm is as simple as the previous ones, the tables are large. Therefore, in a similar way to what was done in the M4 message, I will include those corresponding to the two tastiest patterns, in this case the first and the last. The table of the IDID pattern, in three parts, is as follows:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
	Jump -9-4-9-4						IDID																												
1	Y	T	X	T	Z	M	A	R	Z	I	S	W	Z	S	V	W	Z	H	O	U	Z	H	M	T	Z	O	C	B	U	A	M	W	Y	S	V
2	E	G	R	P	X	L	S	D	X	A	O	J	X	I	U	J	I	D	H	Q	U	D	D	P	X	T	B	A	W	S	L	S	E	I	U
3	T	F	N	Z	W	J	O	C	W	S	U	E	W	H	Q	E	X	G	A	N	X	G	C	Z	Q	Q	A	S	S	O	J	Z	D	H	Q
4	Z	E	M	Y	N	I	N	B	G	O	T	G	V	G	N	G	W	O	Y	M	J	O	B	Y	W	P	S	O	O	T	I	Q	C	G	N
5	R	D	L	Q	V	O	U	A	V	T	P	V	U	R	M	V	V	T	T	B	I	T	A	Q	I	Y	O	R	T	P	E	V	B	R	M
6	D	S	J	N	H	H	T	P	U	P	Y	U	Q	P	L	U	D	P	G	L	H	P	S	N	H	R	T	T	P	N	H	O	T	P	L
7	C	C	I	M	G	G	W	S	Q	Y	J	Q	N	F	J	Q	F	Y	F	J	G	Y	O	M	G	I	Q	V	Z	Y	S	T	A	F	J
8	B	R	H	L	F	F	M	X	B	R	M	N	M	O	I	N	U	R	E	I	F	R	T	L	F	L	P	U	Y	R	O	P	Z	O	I
9	A	K	G	J	E	K	Q	W	D	K	R	I	B	K	O	I	H	K	D	P	E	K	P	J	E	K	Y	Q	Q	K	T	Y	X	K	O
10	P	Z	F	I	D	Y	P	V	N	Z	Q	M	L	Y	H	M	C	Z	S	O	D	Z	Y	I	D	N	R	N	N	H	P	X	W	Y	H
11	S	X	E	H	C	E	Y	U	F	X	I	D	J	E	G	D	E	X	C	H	C	X	R	H	C	H	I	M	M	J	Y	R	V	E	G
12	X	W	D	G	B	T	R	Q	A	H	L	C	I	D	F	C	Q	W	R	A	B	W	L	G	B	J	L	L	L	Z	R	N	U	D	F
13	W	V	K	F	A	Z	K	N	C	W	K	B	P	C	K	B	N	V	K	Y	A	V	K	F	A	Z	K	J	J	M	K	M	Q	C	K
14	V	U	C	E	S	R	V	M	M	V	Z	A	O	B	Y	A	M	F	Z	T	W	F	Z	E	S	X	N	I	I	G	Z	L	N	B	Y
15	U	Q	B	R	O	D	Z	L	L	U	X	S	H	T	E	S	L	U	X	G	S	U	X	R	V	W	H	H	I	X	J	M	T	E	
16	Q	N	A	D	U	C	L	J	J	C	W	O	A	A	T	O	G	Q	W	F	O	Q	F	D	O	V	J	G	G	X	W	I	L	A	T
17	N	M	U	C	T	B	J	I	I	E	N	T	Y	Z	Z	T	J	N	V	E	T	N	H	C	T	M	Z	P	F	W	G	H	J	Z	Z
18	M	B	W	B	P	A	I	O	E	Q	V	P	T	X	R	P	B	A	U	D	M	A	W	B	P	U	X	F	E	V	V	G	S	X	R
19	L	L	S	V	Y	P	H	H	G	H	Y	G	W	D	Y	A	C	Q	S	Q	C	J	V	L	G	W	E	R	U	U	F	I	W	D	
20	J	J	Z	X	J	S	G	G	S	B	G	R	F	V	C	R	S	M	N	C	P	M	E	X	N	F	V	D	D	L	Q	E	H	V	C
21	I	I	Q	A	M	X	F	F	O	D	F	L	E	U	B	L	O	E	M	R	V	E	G	A	Y	E	M	X	C	Q	B	D	G	U	B
22	O	P	V	K	R	W	E	K	T	N	E	K	D	Q	A	K	T	S	B	K	L	S	V	K	U	D	U	K	B	F	D	K	R	Q	A
23	H	O	O	U	Q	V	D	Y	P	M	D	Z	S	N	P	Z	P	B	L	Z	N	B	U	U	J	C	G	C	V	E	N	C	P	N	P
24	G	H	T	W	I	U	C	E	Y	L	C	X	C	M	S	X	Y	L	J	X	Y	L	Q	W	M	B	F	Y	X	D	F	B	F	M	S
25	F	A	P	S	L	Q	B	T	R	J	B	F	R	L	X	F	R	J	I	W	R	J	N	S	R	A	E	W	A	C	A	A	O	L	X
26	K	Y	Y	O	K	N	X	Z	K	F	A	H	K	J	W	H	K	I	P	V	K	I	I	O	K	S	D	Z	K	B	C	U	K	J	W
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
	Jump +9+4+9+4						QVQV																												
1	Y	L	X	V	Z	P	A	H	Z	G	S	Y	Z	W	V	Y	Z	C	O	S	Z	C	M	V	Z	G	C	E	U	U	M	F	Y	W	V
2	E	J	R	X	X	S	S	G	X	B	O	R	X	V	U	R	I	M	H	C	U	M	D	X	X	F	B	D	W	L	L	E	E	V	U
3	T	I	N	A	W	X	O	F	W	D	U	L	W	U	Q	L	X	E	A	R	X	E	C	A	Q	E	A	X	S	Q	J	D	D	U	Q
4	Z	P	M	K	N	W	N	K	G	N	T	K	V	Q	N	K	W	S	Y	K	J	S	B	K	W	D	S	K	O	F	I	K	C	Q	N
5	R	O	L	U	V	V	U	Y	V	M	P	Z	U	N	M	Z	V	B	T	Z	I	B	A	U	I	C	O	C	T	E	E	C	B	N	M
6	D	H	J	W	H	U	T	E	U	L	Y	X	Q	M	L	X	D	L	G	X	H	L	S	W	H	B	T	Y	P	D	H	B	T	M	L
7	C	A	I	S	G	Q	W	T	Q	J	J	F	N	L	J	F	F	J	F	W	G	J	O	S	G	A	Q	W	Z	C	S	A	A	L	J
8	B	Y	H	O	F	N	M	Z	B	F	M	H	M	J	I	H	U	I	E	V	F	I	T	O	F	S	P	Z	Y	B	O	U	Z	J	I
9	A	T	G	T	E	M	Q	R	D	I	R	W	B	S	O	W	H	H	D	U	E	H	P	T	E	O	Y	B	Q	A	T	W	X	S	O
10	P	G	F	P	D	L	P	D	N	A	Q	J	L	I	H	J	C	D	S	Q	D	D	Y	P	D	T	R	A	N	S	P	S	W	I	H
11	S	F	E	Z	C	J	Y	C	F	S	I	E	J	H	G	E	E	G	C	N	C	G	R	Z	C	Q	I	S	M	O	Y	Z	V	H	G
12	X	E	D	Y	B	I	R	B	A	O	L	G	I	G	F	G	Q	O	R	M	B	O	L	Y	B	P	L	O	L	T	R	Q	U	G	F
13	W	D	K	Q	A	O	K	A	C	T	K	V	P	R	K	V	N	T	K	B	A	T	K	Q	A	Y	K	R	J	P	K	V	Q	R	K
14	V	S	C	N	S	H	V	P	M	P	Z	U	O	P	Y	U	M	P	Z	L	W	P	Z	N	S	R	N	T	I	N	Z	O	N	P	Y
15	U	C	B	M	O	G	Z	S	L	Y	X	Q	H	F	E	Q	L	Y	X	J	S	Y	X	M	V	I	H	V	H	Y	X	T	M	F	E
16	Q	R	A	L	U	F	L	X	J	R	W	N	A	O	T	N	G	R	W	I	O	R	F	L	O	L	J	U	G	R	W	P	L	O	T
17	N	K	U	J	T	K	J	W	I	K	N	I	Y	K	Z	I	J	K	V	P	T	K	H	J	T	K	Z	Q	F	K	G	Y	J	K	Z
18	M	Z	W	I	P	Y	I	V	E	Z	V	M	T	Y	R	M	B	Z	U	O	M	Z	W	I	P	N	X	N	E	H	V	X	S	Y	R
19	L	X	S	H	Y	E	H	U	H	X	H	D	G	E	D	D	A	X	Q	H	Q	X	J	H	L	H	W	M	R	J	U	R	I	E	D
20	J	W	Z	G	J	T	G	Q	S	H	G	C	F	D	C	C	S	W	N	A	P	W	E	G	N	J	V	L	D	Z	Q	N	H	D	C
21	I	V	Q	F	M	Z	F	N	O	W	F	B	E	C	B	B	O	V	M	Y	V	V	G	F	Y	Z	M	J	C	M	B	M	G	C	B
22	O	U	V	E	R	R	E	M	T	V	E	A	D	B	A	A	T	F	B	T	L	F	V	E	U	X	U	I	B	G	D	L	R	B	A
23	H	Q	O	R	Q	D	D	L	P	U	D	S	S	T	P	S	P	U	L	G	N	U	U	R	J	W	G	H	V	I	N	J	P	T	P
24	G	N	T	D	I	C	C	J	Y	C	C	O	C	A	S	O	Y	Q	J	F	Y	Q	Q	D	M	V	F	G	X	X	F	I	F	A	S
25	F	M	P	C	L	B	B	I	R	E	B	T	R	Z	X	T	R	N	I	E	R	N	N	C	R	M	E	P	A	W	A	H	O	Z	X
26	K	B	Y	B	K	A	X	O	K	Q	A	P	K	X	W	P	K	A	P	D	K	A	I	B	K	U	D	F	K	V	C	G	K	X	W

36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
Y	Z	G	C	B	J	O	N	B	Z	L	S	H	Z	R	J	E	C	A	S	U	Z	R	W	V	H	H	O	G	P	I
O	U	F	B	T	Z	T	M	T	X	J	Q	G	U	P	I	G	B	S	O	Q	I	P	S	D	J	D	H	I	U	H
W	X	E	A	A	X	Q	L	A	W	I	V	F	X	F	H	V	A	O	U	D	X	F	O	F	Z	G	A	X	Q	G
V	J	Z	S	Z	W	P	J	Z	V	H	O	E	J	E	T	I	U	T	T	N	W	E	T	U	M	O	Y	W	N	P
U	I	R	O	X	E	Y	S	X	U	G	X	D	I	D	G	D	W	P	P	M	V	D	M	H	G	T	T	V	M	F
Q	H	D	T	W	G	R	I	W	E	F	N	C	H	C	F	F	S	Y	Y	L	D	C	Q	C	I	P	G	U	L	E
N	G	P	Q	V	V	I	H	V	Q	E	U	Z	G	O	E	U	Z	R	J	O	F	O	P	E	X	Y	F	L	J	D
M	F	X	P	U	I	L	G	U	N	D	T	B	F	B	Z	Q	Q	L	M	A	U	B	V	Q	W	R	E	Q	I	X
L	E	K	Y	Q	D	K	R	Q	M	K	P	A	E	K	R	N	V	K	R	J	H	K	L	N	V	K	D	F	H	K
J	D	C	R	N	F	N	P	N	S	C	Y	S	D	Z	D	M	O	Z	Q	C	C	Z	N	M	U	Z	S	E	G	C
I	C	B	I	M	U	H	F	M	B	B	R	Q	C	X	P	H	T	X	I	T	E	X	Y	L	L	X	C	D	F	Y
H	B	A	L	L	Q	J	O	L	L	A	W	V	B	W	X	L	P	F	L	S	Q	W	R	G	Q	W	R	C	Y	W
T	A	S	K	J	N	Z	K	J	D	U	K	O	A	V	K	C	Y	H	K	I	N	V	K	J	F	V	K	B	E	Z
G	W	Y	N	S	M	X	Y	S	O	W	M	X	W	U	C	B	X	W	Z	H	M	U	Z	B	E	F	Z	A	D	B
F	S	O	H	I	H	W	E	I	A	S	L	N	S	Q	B	A	R	J	X	G	L	Q	U	A	D	U	X	S	C	A
E	O	W	J	H	L	V	D	H	J	Z	J	U	O	N	A	S	N	E	W	F	G	N	X	S	C	Q	W	O	W	S
Z	T	V	Z	G	C	M	C	G	I	Q	I	T	T	M	S	O	M	G	N	B	J	M	J	O	B	N	V	T	Z	O
R	M	U	X	R	B	U	B	R	H	V	H	P	M	L	Y	T	L	V	V	E	B	L	I	T	A	A	U	P	B	R
D	Q	Q	W	P	A	G	T	P	G	O	G	Y	Q	A	O	P	J	U	H	P	A	A	H	P	S	C	Q	N	R	T
P	P	N	V	F	S	F	A	F	C	T	F	R	P	J	W	Y	I	Q	G	Y	S	J	G	Y	O	M	N	Y	V	V
X	V	M	M	O	O	E	Z	O	F	P	E	W	V	I	V	R	H	N	F	R	O	I	F	R	T	E	M	R	X	U
K	L	L	U	K	T	D	X	K	T	Y	D	K	L	H	U	K	G	I	E	K	T	H	E	K	P	S	B	K	A	Q
C	N	J	G	Y	P	C	W	Y	P	X	C	M	N	Y	Q	J	F	M	D	Z	P	Y	D	Z	N	B	L	H	S	N
B	Y	I	F	E	Y	B	V	E	Y	R	Z	L	Y	T	N	Z	E	D	C	X	Y	T	C	I	Y	L	J	J	O	M
A	R	H	E	D	R	A	U	D	R	N	B	J	R	G	M	X	D	C	B	W	R	G	B	X	R	J	I	Z	T	L
S	K	T	D	C	K	S	Q	C	K	M	A	I	K	S	L	W	K	B	A	V	K	S	A	W	K	I	P	M	K	J
36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
D	Z	Q	C	P	J	G	N	P	Z	O	S	Y	Z	A	J	P	C	U	S	P	Z	A	W	P	H	C	O	N	P	T
P	U	N	B	F	Z	F	M	F	X	T	Q	R	U	J	I	Y	B	Q	O	Y	I	J	S	Y	J	M	H	Y	U	V
X	X	M	A	O	X	E	L	O	W	P	V	W	X	I	H	R	A	N	U	R	X	I	O	R	Z	E	A	R	Q	U
K	J	L	S	K	W	D	J	K	V	Y	O	K	J	H	T	K	U	I	T	K	W	H	T	K	M	S	Y	K	N	Q
C	I	J	O	Y	E	C	S	Y	U	X	X	M	I	Y	G	J	W	M	P	Z	V	Y	M	Z	G	B	T	H	M	N
B	H	I	T	E	G	B	I	E	E	R	N	L	H	T	F	Z	S	D	Y	X	D	T	Q	I	I	L	G	J	L	M
A	G	H	Q	D	V	A	H	D	Q	N	U	J	G	G	E	X	Z	C	J	W	F	G	P	X	X	J	F	Z	J	L
S	F	T	P	C	I	S	G	C	N	M	T	I	F	S	Z	W	Q	B	M	V	U	S	V	W	W	I	E	M	I	J
Y	E	G	Y	B	D	O	R	B	M	L	P	H	E	R	R	E	V	A	R	U	H	R	L	V	V	H	D	G	H	I
O	D	F	R	T	F	T	P	T	S	J	Y	G	D	P	D	G	O	S	Q	Q	C	P	N	D	U	D	S	I	G	H
W	C	E	I	A	U	Q	F	A	B	I	R	F	C	F	P	V	T	O	I	D	E	F	Y	F	L	G	C	X	F	G
V	B	Z	L	Z	Q	P	O	Z	L	H	W	E	B	E	X	I	P	T	L	N	Q	E	R	U	Q	O	R	W	Y	P
U	A	R	K	X	N	Y	K	X	D	G	K	D	A	D	K	D	Y	P	K	M	N	D	K	H	F	T	K	V	E	F
Q	W	D	N	W	M	R	Y	W	O	F	M	C	W	C	C	F	X	Y	Z	L	M	C	Z	C	E	P	Z	U	D	E
N	S	P	H	V	H	I	E	V	A	E	L	Z	S	O	B	U	R	R	X	O	L	O	U	E	D	Y	X	L	C	D
M	O	X	J	U	L	D	U	J	D	J	B	O	B	A	Q	N	L	W	A	G	B	X	Q	C	R	W	Q	W	X	X
L	T	K	Z	Q	C	K	C	Q	I	K	I	A	T	K	S	N	M	K	N	J	K	J	N	B	K	V	F	Z	K	K
J	M	C	X	N	B	N	B	N	H	C	H	S	M	Z	Y	M	L	Z	V	C	B	Z	I	M	A	Z	U	E	B	C
I	Q	B	W	M	A	H	T	M	G	B	G	Q	X	O	H	J	X	H	T	A	X	H	L	S	X	Q	D	R	Y	Y
H	P	A	V	L	S	J	A	L	C	A	F	V	P	W	W	L	I	F	G	S	S	W	G	G	O	W	N	C	V	W
T	V	S	M	J	O	Z	Z	J	F	U	E	O	V	V	V	C	H	H	F	I	O	V	F	J	T	V	M	B	X	Z
G	L	Y	U	S	T	X	X	S	T	W	D	X	L	U	U	B	G	W	E	H	T	U	E	B	P	F	B	A	A	B
F	N	O	G	I	P	W	W	I	P	S	C	N	N	Q	Q	A	F	J	D	G	P	Q	D	A	N	U	L	S	S	A
E	Y	W	F	H	Y	V	V	H	Y	Z	Z	U	Y	N	N	S	E	E	C	F	Y	N	C	S	Y	Q	J	O	O	S
Z	R	V	E	G	R	M	U	G	R	Q	B	T	R	M	M	O	D	G	B	B	R	M	B	O	R	N	I	T	T	O
R	K	U	D	R	K	U	Q	R	K	V	A	P	K	L	L	T	K	V	A	E	K	L	A	T	K	A	P	P	K	R

67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R
S	Y	C	A	S	F	C	L	W	V	Z	L	S	I	C	B	I	U	Z	E	X	V	Z	P	Y	V	Z	L	I	U	U
V	O	B	S	Q	U	Y	J	S	U	X	J	Q	H	B	A	B	Q	I	H	R	H	X	Y	T	H	X	J	B	Q	W
O	W	A	O	V	Q	W	I	O	Q	W	I	V	G	A	S	P	D	X	S	N	G	H	J	G	G	W	I	P	N	S
T	V	S	T	O	N	Z	H	T	B	V	H	O	F	S	O	O	N	W	O	M	F	W	M	S	F	V	H	O	M	O
P	U	Y	P	X	A	B	G	M	D	U	G	X	E	Y	T	H	M	V	T	L	E	V	R	R	E	U	G	H	L	T
L	Q	O	Y	N	C	A	F	Q	N	Q	F	N	D	O	P	G	L	D	P	J	D	U	Q	P	D	E	F	G	A	P
N	N	W	M	U	M	S	E	P	F	D	E	U	C	W	Y	F	O	F	Y	I	C	C	I	F	C	Q	E	F	J	Z
Y	M	V	R	T	E	O	D	V	A	N	D	T	B	V	R	E	A	U	R	H	B	E	L	E	B	N	D	E	I	Y
U	L	U	K	P	S	R	K	L	C	M	K	P	X	U	K	A	J	H	K	G	A	Q	K	D	A	M	K	A	H	Q
J	J	Q	Z	Y	B	T	C	N	M	L	C	Y	A	Q	J	D	C	C	Z	F	S	G	Z	C	S	S	C	D	Y	N
M	I	N	G	R	L	V	B	Y	L	O	B	R	S	N	Z	Y	T	E	X	E	O	B	X	O	O	B	B	Y	T	M
R	H	M	I	W	J	U	A	R	J	A	A	W	O	M	X	R	S	Q	W	D	U	D	W	B	U	L	A	R	G	L
K	T	L	X	K	I	Q	U	K	I	J	U	K	N	L	W	K	I	N	G	K	T	N	N	K	T	D	U	K	S	J
Z	G	J	L	M	H	N	W	Z	E	C	W	M	U	J	E	Z	H	M	V	C	P	M	V	Z	P	O	W	Z	R	I
X	F	I	F	L	D	M	S	U	H	T	S	L	T	I	G	X	G	L	U	B	Y	L	H	X	Y	A	S	X	P	H
Q	E	H	H	J	G	L	Z	X	S	S	Z	J	W	H	V	W	F	G	Q	A	J	J	G	W	J	J	Z	W	F	G
W	Z	T	W	I	O	J	Q	J	O	I	Q	I	M	T	I	V	B	J	B	U	M	F	F	V	M	I	Q	V	E	F
I	R	G	V	H	T	I	V	I	T	H	V	H	Q	G	D	U	E	B	D	W	R	I	E	U	R	H	V	U	D	E
H	D	F	U	G	P	H	O	H	P	G	O	G	P	F	F	Q	P	A	N	S	Q	A	D	Q	Q	G	O	Q	C	R
G	P	E	Q	F	Y	G	T	G	Y	F	T	F	Y	E	U	N	Y	S	F	Z	I	S	C	N	I	C	T	N	O	D
F	X	Z	J	E	R	P	P	F	R	B	P	E	R	Z	Q	C	R	O	A	Q	L	O	B	M	L	F	P	C	B	C
E	K	R	N	D	K	F	Y	E	K	E	Y	D	K	R	N	M	K	T	C	V	K	T	A	L	K	T	Y	M	K	B
D	C	D	E	C	Z	E	X	D	Z	P	X	C	V	D	M	L	Z	P	M	O	Z	P	S	A	Z	P	X	L	Z	V
C	B	P	D	Z	X	D	R	C	X	Y	R	Z	Z	P	H	J	X	Y	L	T	X	Y	O	J	X	Y	R	J	X	X
B	A	X	C	B	W	X	N	B	W	R	N	B	L	X	L	T	W	R	J	P	W	R	U	I	W	R	N	T	W	A
A	S	K	B	A	V	K	M	A	G	K	M	A	J	K	C	S	V	K	I	Y	N	K	T	H	N	K	M	S	V	K
67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R
S	D	C	U	S	P	C	O	W	P	Z	O	S	P	C	F	I	P	Z	N	X	Q	Z	D	Y	Q	Z	O	I	C	U
V	P	B	Q	Q	Y	Y	T	S	Y	X	T	Q	Y	B	U	B	Y	I	F	R	I	X	C	T	I	X	T	B	O	W
O	X	A	J	V	R	W	P	O	R	W	P	V	R	A	Q	P	R	X	A	N	L	H	B	G	L	W	P	P	B	S
T	K	S	N	O	K	Z	Y	T	K	V	Y	O	K	S	N	O	K	W	C	M	K	W	A	S	K	V	Y	O	K	O
P	C	Y	E	X	Z	B	X	M	Z	U	X	X	V	Y	M	H	Z	V	M	L	Z	V	S	R	Z	U	X	H	Z	T
L	B	O	D	N	X	A	R	Q	X	Q	R	N	Z	O	H	G	X	D	L	J	X	U	O	P	X	E	R	G	X	P
N	A	W	C	U	W	S	N	P	W	D	N	U	L	W	L	F	W	F	J	I	W	C	U	F	W	Q	N	F	W	Z
Y	S	V	B	T	V	O	M	V	G	N	M	T	J	V	C	E	V	U	I	H	N	E	T	E	N	N	M	E	V	Y
U	Y	U	A	P	F	R	L	L	V	M	L	P	I	U	B	A	U	H	E	G	V	Q	P	D	V	M	L	A	U	Q
J	O	Q	S	Y	U	T	J	N	U	L	J	Y	H	Q	A	D	Q	C	H	F	H	G	Y	C	H	S	J	D	Q	N
M	W	N	O	R	Q	V	I	Y	Q	O	I	R	G	N	S	Y	D	E	S	E	G	B	J	O	G	B	I	Y	N	M
R	V	M	T	W	N	U	H	R	B	A	H	W	F	M	O	R	N	Q	O	D	F	D	M	B	F	L	H	R	M	L
K	U	L	P	K	A	Q	G	K	D	J	G	K	E	L	T	K	M	N	T	K	E	N	R	K	E	D	G	K	L	J
Z	Q	J	Y	M	C	N	F	Z	N	C	F	M	D	J	P	Z	L	M	P	C	D	M	Q	Z	D	O	F	Z	A	I
X	N	I	M	L	M	M	E	U	F	T	E	L	C	I	Y	X	O	L	Y	B	C	L	I	X	C	A	E	X	J	H
Q	M	H	R	J	E	L	D	X	A	S	D	J	B	H	R	W	A	G	R	A	B	J	L	W	B	J	D	W	I	G
W	L	T	K	I	S	J	K	J	C	I	K	I	X	T	K	V	J	J	K	U	A	F	K	V	A	I	K	V	H	F
I	J	G	Z	H	B	I	C	I	M	H	C	H	A	G	J	U	C	B	Z	W	S	I	Z	U	S	H	C	U	Y	E
H	I	F	G	G	L	H	B	H	L	G	B	G	S	F	Z	Q	T	A	X	S	O	A	X	Q	O	G	B	Q	T	R
G	H	E	I	F	J	G	A	G	J	F	A	F	O	E	X	N	S	S	W	Z	U	S	W	N	U	C	A	N	G	D
F	T	Z	X	E	I	P	U	F	I	B	U	E	N	Z	W	C	I	O	G	Q	T	O	N	M	T	F	U	C	S	C
E	G	R	L	D	H	F	W	E	E	E	W	D	U	R	E	M	H	T	V	V	P	T	V	L	P	T	W	M	R	B
D	F	D	F	C	D	E	S	D	H	P	S	C	T	D	G	L	G	P	U	O	Y	P	H	A	Y	P	S	L	P	V
C	E	P	H	Z	G	D	Z	C	S	Y	Z	Z	W	P	V	J	F	Y	Q	T	J	Y	G	J	J	Y	Z	J	F	X
B	Z	X	W	B	O	X	Q	B	O	R	Q	B	M	X	I	T	B	R	B	P	M	R	F	I	M	R	Q	T	E	A
A	R	K	V	A	T	K	V	A	T	K	V	A	Q	K	D	S	E	K	D	Y	R	K	E	H	R	K	V	S	D	K

IDID pattern #17 BAUD, #26 TRANSPSW + #34 ED (80% of TRANSPOSED? Also the O is shift -1 of the S in the Kryptos alphabet), #31 HOTPL (80% of HOTEL?), #36 TASK , #37 BALL, #58 QUAD, #59 BORN, #81 SOON, #83 KING and #89 TALK.

DIDI pattern #31 HRTEL (80% of HOTEL? In the same place #31 as in the previous pattern) and #37 ELY + #40 OIE (ELYOIE the exact key of BERLIN! Finally Kryptos confirms the key, and with NYPVTT the right place of BERLIN).

IIDD pattern #67 LIN (in the right place!)

DDII pattern #26 YORK, #37 CITE, #40 OIE.

IDDI pattern #23 PEEK, #27 GAVE, #31 SEAL (this appear later), #34 BALL, #37 IRON, #41 GRIME, #88 TELE and #89 PHASP (80% PHASE?).

The table of the DIID pattern is:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
	Jump-4-9-9-4					DIID																													
1	Y	S	X	T	Z	H	A	R	Z	P	S	W	Z	P	V	W	Z	P	O	U	Z	P	M	T	Z	R	C	B	U	N	M	W	Y	P	V
2	E	C	R	P	X	G	S	D	X	Y	O	J	X	F	U	J	I	Y	H	Q	U	Y	D	P	X	I	B	A	W	Y	L	S	E	F	U
3	T	R	N	Z	W	F	O	C	W	R	U	E	W	O	Q	E	X	R	A	N	X	R	C	Z	Q	L	A	S	S	R	J	Z	D	O	Q
4	Z	K	M	Y	N	K	N	B	G	K	T	G	V	K	N	G	W	K	Y	M	J	K	B	Y	W	K	S	O	O	K	I	Q	C	K	N
5	R	Z	L	Q	V	Y	U	A	V	Z	P	V	U	Y	M	V	V	Z	T	B	I	Z	A	Q	I	N	O	R	T	H	E	V	B	Y	M
6	D	X	J	N	H	E	T	P	U	X	Y	U	Q	E	L	U	D	X	G	L	H	X	S	N	H	H	T	T	P	J	H	O	T	E	L
7	C	W	I	M	G	T	W	S	Q	H	J	Q	N	D	J	Q	F	W	F	J	G	W	O	M	G	J	Q	V	Z	Z	S	T	A	D	J
8	B	V	H	L	F	Z	M	X	B	W	M	N	M	C	I	N	U	V	E	I	F	V	T	L	F	Z	P	U	Y	M	O	P	Z	C	I
9	A	U	G	J	E	R	Q	W	D	V	R	I	B	B	O	I	H	F	D	P	E	F	P	J	E	X	Y	Q	Q	G	T	Y	X	B	O
10	P	Q	F	I	D	D	P	V	N	U	Q	M	L	T	H	M	C	U	S	O	D	U	Y	I	D	W	R	N	N	I	P	X	W	T	H
11	S	N	E	H	C	C	Y	U	F	C	I	D	J	A	G	D	E	Q	C	H	C	Q	R	H	C	V	I	M	M	X	Y	R	V	A	G
12	X	M	D	G	B	B	R	Q	A	E	L	C	I	Z	F	C	Q	N	R	A	B	N	L	G	B	M	L	L	L	W	R	N	U	Z	F
13	W	B	K	F	A	A	K	N	C	Q	K	B	P	X	K	B	N	A	K	Y	A	A	K	F	A	U	K	J	J	V	K	M	Q	X	K
14	V	L	C	E	S	P	V	M	M	G	Z	A	O	W	Y	A	M	C	Z	T	W	C	Z	E	S	G	N	I	I	U	Z	L	N	W	Y
15	U	J	B	R	O	S	Z	L	L	B	X	S	H	V	E	S	L	M	X	G	S	M	X	R	V	F	H	H	L	X	J	M	V	E	
16	Q	I	A	D	U	X	L	J	J	D	W	O	A	U	T	O	G	E	W	F	O	E	F	D	O	E	J	G	G	Q	W	I	L	U	T
17	N	P	U	C	T	W	J	I	I	N	T	Y	Q	Z	T	J	S	V	E	T	S	H	C	T	D	Z	P	F	F	G	H	J	Q	Z	
18	M	O	W	B	P	V	I	O	E	M	V	P	T	N	R	P	B	B	U	D	M	B	W	B	P	C	X	F	E	E	V	G	S	N	R
19	L	H	S	V	Y	U	H	H	L	H	Y	G	M	D	Y	A	L	Q	S	Q	L	J	V	L	B	W	E	R	D	U	F	I	M	D	
20	J	A	Z	X	J	Q	G	G	S	J	G	R	F	L	C	R	S	J	N	C	P	J	E	X	N	A	V	D	D	C	Q	E	H	L	C
21	I	Y	Q	A	M	N	F	F	O	F	F	L	E	J	B	L	O	I	M	R	V	I	G	A	Y	S	M	X	C	B	B	D	G	J	B
22	O	T	V	K	R	M	E	K	T	I	E	K	D	S	A	K	T	H	B	K	L	H	V	K	U	O	U	K	B	A	D	K	R	S	A
23	H	G	O	U	Q	L	D	Y	P	A	D	Z	S	I	P	Z	P	D	L	Z	N	D	U	U	J	T	G	C	V	S	N	C	P	I	P
24	G	F	T	W	I	J	C	E	Y	S	C	X	C	H	S	X	Y	G	J	X	Y	G	Q	W	M	Q	F	Y	X	O	F	B	F	H	S
25	F	E	P	S	L	I	B	T	R	O	B	F	R	G	X	F	R	O	I	W	R	O	N	S	R	P	E	W	A	T	A	A	O	G	X
26	K	D	Y	O	K	O	X	Z	K	T	A	H	K	R	W	H	K	T	P	V	K	T	I	O	K	Y	D	Z	K	P	C	U	K	R	W
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
	Jump+4+9+9+4					VQQV																													
1	Y	U	X	V	Z	R	A	H	Z	V	S	Y	Z	B	V	Y	Z	F	O	S	Z	F	M	V	Z	X	C	E	U	G	M	F	Y	B	V
2	E	Q	R	X	X	D	S	G	X	U	O	R	X	T	U	R	I	U	H	C	U	U	D	X	X	W	B	D	W	I	L	E	E	T	U
3	T	N	N	A	W	C	O	F	W	C	U	L	W	A	Q	L	X	Q	A	R	X	Q	C	A	Q	V	A	X	S	X	J	D	D	A	Q
4	Z	M	M	K	N	B	N	K	G	E	T	K	V	Z	N	K	W	N	Y	K	J	N	B	K	W	M	S	K	O	W	I	K	C	Z	N
5	R	B	L	U	V	A	U	Y	V	Q	P	Z	U	X	M	Z	V	A	T	Z	I	A	A	U	I	U	O	C	T	V	E	C	B	X	M
6	D	L	J	W	H	P	T	E	U	G	Y	X	Q	W	L	X	D	C	G	X	H	C	S	W	H	G	T	Y	P	U	H	B	T	W	L
7	C	J	I	S	G	S	W	T	Q	B	J	F	N	V	J	F	F	M	F	W	G	M	O	S	G	F	Q	W	Z	L	S	A	A	V	J
8	B	I	H	O	F	X	M	Z	B	D	M	H	M	U	I	H	U	E	E	V	F	E	T	O	F	E	P	Z	Y	Q	O	U	Z	U	I
9	A	P	G	T	E	W	Q	R	D	N	R	W	B	Q	O	W	H	S	D	U	E	S	P	T	E	D	Y	B	Q	F	T	W	X	Q	O
10	P	O	F	P	D	V	P	D	N	M	Q	J	L	N	H	J	C	B	S	Q	D	B	Y	P	D	C	R	A	N	E	P	S	W	N	H
11	S	H	E	Z	C	U	Y	C	F	L	I	E	J	M	G	E	E	L	C	N	C	L	R	Z	C	B	I	S	M	D	Y	Z	V	M	G
12	X	A	D	Y	B	Q	R	B	A	J	L	G	I	L	F	G	Q	J	R	M	B	J	L	Y	B	A	L	O	L	C	R	Q	U	L	F
13	W	Y	K	Q	A	N	K	A	C	F	K	V	P	J	K	V	N	I	K	B	A	I	K	Q	A	S	K	R	J	B	K	V	Q	J	K
14	V	T	C	N	S	M	V	P	M	I	Z	U	O	S	Y	U	M	H	Z	L	W	H	Z	N	S	O	N	T	I	A	Z	O	N	S	Y
15	U	G	B	M	O	L	Z	S	L	A	X	Q	H	I	E	Q	L	D	X	J	S	D	X	M	V	T	H	V	H	S	X	T	M	I	E
16	Q	F	A	L	U	J	L	X	J	S	W	N	A	H	T	N	G	G	W	I	O	G	F	L	O	Q	J	U	G	O	W	P	L	H	T
17	N	E	U	J	T	I	J	W	I	O	N	I	Y	G	Z	I	J	O	V	P	T	O	H	J	T	P	Z	Q	F	T	G	Y	J	G	Z
18	M	D	W	I	P	O	I	V	E	T	V	M	T	R	R	M	B	T	U	O	M	T	W	I	P	Y	X	N	E	P	V	X	S	R	R
19	L	S	S	H	Y	H	H	U	H	P	H	D	G	P	D	D	A	P	Q	H	Q	P	J	H	L	R	W	M	R	N	U	R	I	P	D
20	J	C	Z	G	J	G	G	Q	S	Y	G	C	F	F	C	C	S	Y	N	A	P	Y	E	G	N	I	V	L	D	Y	Q	N	H	F	C
21	I	R	Q	F	M	F	F	N	O	R	F	B	E	O	B	B	O	R	M	Y	V	R	G	F	Y	L	M	J	C	R	B	M	G	O	B
22	O	K	V	E	R	K	E	M	T	K	E	A	D	K	A	A	T	K	B	T	L	K	V	E	U	K	I	B	K	D	L	R	K	A	
23	H	Z	O	R	Q	Y	D	L	P	Z	D	S	S	Y	P	S	P	Z	L	G	N	Z	U	R	J	N	G	H	V	H	N	J	P	Y	P
24	G	X	T	D	I	E	C	J	Y	X	C	O	C	E	S	O	Y	X	J	F	Y	X	Q	D	M	H	F	G	X	J	F	I	F	E	S
25	F	W	P	C	L	T	B	I	R	H	B	T	R	D	X	T	R	W	I	E	R	W	N	C	R	J	E	P	A	Z	A	H	O	D	X
26	K	V	Y	B	K	Z	X	O	K	W	A	P	K	C	W	P	K	V	P	D	K	V	I	B	K	Z	D	F	K	M	C	G	K	C	W

36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
Y	Z	D	C	B	J	R	N	B	Z	F	S	H	Z	C	J	E	C	Y	S	U	Z	C	W	V	H	P	O	G	P	E
O	U	P	B	T	Z	I	M	T	X	E	Q	G	U	O	I	G	B	R	O	Q	I	O	S	D	J	Y	H	I	U	D
W	X	X	A	A	X	L	L	A	W	D	V	F	X	B	H	V	A	L	U	D	X	B	O	F	Z	R	A	X	Q	X
V	J	K	S	Z	W	K	J	Z	V	K	O	E	J	K	T	I	U	K	T	N	W	K	T	U	M	K	Y	W	N	K
U	I	C	O	X	E	N	S	X	U	C	X	D	I	Z	G	D	W	Z	P	M	V	Z	M	H	G	Z	T	V	M	C
Q	H	B	T	W	G	H	I	W	E	B	N	C	H	X	F	F	S	X	Y	L	D	X	Q	C	I	X	G	U	L	Y
N	G	A	Q	V	V	J	H	V	Q	A	U	Z	G	W	E	U	Z	F	J	O	F	W	P	E	X	W	F	L	J	W
M	F	S	P	U	I	Z	G	U	N	A	T	B	F	V	Z	Q	Q	H	M	A	U	V	V	Q	W	V	E	Q	I	Z
L	E	Y	Y	Q	D	X	R	Q	M	W	P	A	E	U	R	N	V	W	R	J	H	U	L	N	V	F	D	F	H	B
J	D	O	R	N	F	W	P	N	S	S	Y	S	D	Q	D	M	O	J	Q	C	C	Q	N	M	U	U	S	E	G	A
I	C	W	I	M	U	V	F	M	B	Z	R	Q	C	N	P	H	T	E	I	T	E	N	Y	L	L	Q	C	D	F	S
H	B	V	L	L	Q	M	O	L	L	Q	W	V	B	M	X	L	P	G	L	S	Q	M	R	G	Q	N	R	C	Y	O
T	A	U	K	J	N	U	K	J	D	V	K	O	A	L	K	C	Y	V	K	I	N	L	K	J	F	A	K	B	E	R
G	W	Q	N	S	M	G	Y	S	O	O	M	X	W	A	C	B	X	U	Z	H	M	A	Z	B	E	C	Z	A	D	T
F	S	N	H	I	H	F	E	I	A	T	L	N	S	J	B	A	R	Q	X	G	L	J	U	A	D	M	X	S	C	V
E	O	M	J	H	L	E	D	H	J	P	J	U	O	I	A	S	N	N	W	F	G	I	X	S	C	E	W	O	W	U
Z	T	L	Z	G	C	D	C	G	I	Y	I	T	T	H	S	O	M	I	N	B	J	H	J	O	B	S	V	T	Z	Q
R	M	J	X	R	B	C	B	R	H	X	H	P	M	Y	Y	T	L	M	V	E	B	Y	I	T	A	B	U	P	B	N
D	Q	I	W	P	A	B	T	P	G	R	G	Y	Q	T	O	P	J	D	H	P	A	T	H	P	S	L	Q	N	R	M
P	P	H	V	F	S	A	A	F	C	N	F	R	P	G	W	Y	I	C	G	Y	S	G	G	Y	O	J	N	Y	V	L
X	V	T	M	O	O	S	Z	O	F	M	E	W	V	S	V	R	H	B	F	R	O	S	F	R	T	I	M	R	X	J
K	L	G	U	K	T	O	X	K	T	L	D	K	L	R	U	K	G	A	E	K	T	R	E	K	P	H	B	K	A	I
C	N	F	G	Y	P	T	W	Y	P	J	C	M	N	P	Q	J	F	S	D	Z	P	P	D	Z	N	D	L	H	S	H
B	Y	E	F	E	Y	Q	V	E	Y	I	Z	L	Y	F	N	Z	E	O	C	X	Y	F	C	I	Y	G	J	J	O	G
A	R	Z	E	D	R	P	U	D	R	H	B	J	R	E	M	X	D	T	B	W	R	E	B	X	R	O	I	Z	T	P
S	K	R	D	C	K	Y	Q	C	K	G	A	I	K	D	L	W	K	P	A	V	K	D	A	W	K	T	P	M	K	F
36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
D	Z	Y	C	P	J	X	N	P	Z	W	S	Y	Z	U	J	P	C	W	S	P	Z	U	W	P	H	F	O	N	P	B
P	U	O	B	F	Z	W	M	F	X	S	Q	R	U	Q	I	Y	B	J	O	Y	I	Q	S	Y	J	U	H	Y	U	A
X	X	W	A	O	X	V	L	O	W	Z	V	W	X	N	H	R	A	E	U	R	X	N	O	R	Z	Q	A	R	Q	S
K	J	V	S	K	W	M	J	K	V	Q	O	K	J	M	T	K	U	G	T	K	W	M	T	K	M	N	Y	K	N	O
C	I	U	O	Y	E	U	S	Y	U	V	X	M	I	L	G	J	W	V	P	Z	V	L	M	Z	G	A	T	H	M	R
B	H	Q	T	E	G	G	I	E	E	O	N	L	H	A	F	Z	S	U	Y	X	D	A	Q	I	I	C	G	J	L	T
A	G	N	Q	D	V	F	H	D	Q	T	U	J	G	J	E	X	Z	Q	J	W	F	J	P	X	X	M	F	Z	J	V
S	F	M	P	C	I	E	G	C	N	P	T	I	F	I	Z	W	Q	N	M	V	U	I	V	W	W	E	E	M	I	U
Y	E	L	Y	B	D	D	R	B	M	Y	P	H	E	H	R	E	V	I	R	U	H	H	L	V	V	S	D	G	H	Q
O	D	J	R	T	F	C	P	T	S	X	Y	G	D	Y	D	G	O	M	Q	Q	C	Y	N	D	U	B	S	I	G	N
W	C	I	I	A	U	B	F	A	B	R	R	F	C	T	P	V	T	D	I	D	E	T	Y	F	L	L	C	X	F	M
V	B	H	L	Z	Q	A	O	Z	L	N	W	E	B	G	X	I	P	C	L	N	Q	G	R	U	Q	J	R	W	Y	L
U	A	T	K	X	N	S	K	X	D	M	K	D	A	S	K	D	Y	B	K	M	N	S	K	H	F	I	K	V	E	J
Q	W	G	N	W	M	O	Y	W	O	L	M	C	W	R	C	F	X	A	Z	L	M	R	Z	C	E	H	Z	U	D	I
N	S	F	H	V	H	T	E	V	A	J	L	Z	S	P	B	U	R	S	X	O	L	P	U	E	D	D	X	L	C	H
M	O	E	J	U	L	Q	D	U	J	I	J	B	O	F	A	Q	N	O	W	A	G	F	X	Q	C	G	W	Q	W	G
L	T	Z	Z	Q	C	P	C	Q	I	H	I	A	T	E	S	N	M	T	N	J	J	E	J	N	B	O	V	F	Z	P
J	M	R	X	N	B	Y	B	N	H	G	H	S	M	D	Y	M	L	P	V	C	B	D	I	M	A	T	U	E	B	F
I	Q	D	W	M	A	R	T	M	G	F	G	Q	Q	C	O	H	J	Y	H	T	A	C	H	L	S	P	Q	D	R	E
H	P	P	V	L	S	I	A	L	C	E	F	V	P	O	W	L	I	R	G	S	S	O	G	G	O	Y	N	C	V	D
T	V	X	M	J	O	L	Z	J	F	D	E	O	V	B	V	C	H	L	F	I	O	B	F	J	T	R	M	B	X	X
G	L	K	U	S	T	K	X	S	T	K	D	X	L	K	U	B	G	K	E	H	T	K	E	B	P	K	B	A	A	K
F	N	C	G	I	P	N	W	I	P	C	C	N	N	Z	Q	A	F	Z	D	G	P	Z	D	A	N	Z	L	S	S	C
E	Y	B	F	H	Y	H	V	H	Y	B	Z	U	Y	X	N	S	E	X	C	F	Y	X	C	S	Y	X	J	O	O	Y
Z	R	A	E	G	R	J	U	G	R	A	B	T	R	W	M	O	D	F	B	B	R	W	B	O	R	W	I	T	T	W
R	K	S	D	R	K	Z	Q	R	K	U	A	P	K	V	L	T	K	H	A	E	K	V	A	T	K	V	P	P	K	Z

67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R
S	Y	C	Y	S	F	C	F	W	V	Z	F	S	I	C	P	I	U	Z	P	X	V	Z	Q	Y	V	Z	F	I	U	U
V	O	B	M	Q	U	Y	E	S	U	X	E	Q	H	B	Y	B	Q	I	Y	R	H	X	I	T	H	X	E	B	Q	W
O	W	A	R	V	Q	W	D	O	Q	W	D	V	G	A	R	P	D	X	R	N	G	H	L	G	G	W	D	P	N	S
T	V	S	K	O	N	Z	K	T	B	V	K	O	F	S	K	O	N	W	K	M	F	W	K	S	F	V	K	O	M	O
P	U	Y	Z	X	A	B	C	M	D	U	C	X	E	Y	J	H	M	V	Z	L	E	V	Z	R	E	U	C	H	L	T
L	Q	O	G	N	C	A	B	Q	N	Q	B	N	D	O	Z	G	L	D	X	J	D	U	X	P	D	E	B	G	A	P
N	N	W	I	U	M	S	A	P	F	D	A	U	C	W	X	F	O	F	W	I	C	C	W	F	C	Q	A	F	J	Z
Y	M	V	X	T	E	O	U	V	A	N	U	T	B	V	W	E	A	U	G	H	B	E	N	E	B	N	U	E	I	Y
U	L	U	L	P	S	R	W	L	C	M	W	P	X	U	E	A	J	H	V	G	A	Q	V	D	A	M	W	A	H	Q
J	J	Q	F	Y	B	T	S	N	M	L	S	Y	A	Q	G	D	C	C	U	F	S	G	H	C	S	S	S	D	Y	N
M	I	N	H	R	L	V	Z	Y	L	O	Z	R	S	N	V	Y	T	E	Q	E	O	B	G	O	O	B	Z	Y	T	M
R	H	M	W	W	J	U	Q	R	J	A	Q	W	O	M	I	R	S	Q	B	D	U	D	F	B	U	L	Q	R	G	L
K	T	L	V	K	I	Q	V	K	I	J	V	K	N	L	D	K	I	N	D	K	T	N	E	K	T	D	V	K	S	J
Z	G	J	U	M	H	N	O	Z	E	C	O	M	U	J	F	Z	H	M	N	C	P	M	D	Z	P	O	O	Z	R	I
X	F	I	Q	L	D	M	T	U	H	T	T	L	T	I	U	X	G	L	F	B	Y	L	C	X	Y	A	T	X	P	H
Q	E	H	J	J	G	L	P	X	S	S	P	J	W	H	Q	W	F	G	A	A	J	J	B	W	J	J	P	W	F	G
W	Z	T	N	I	O	J	Y	J	O	I	Y	I	M	T	N	V	B	J	C	U	M	F	A	V	M	I	Y	V	E	F
I	R	G	E	H	T	I	X	I	T	H	X	H	Q	G	M	U	E	B	M	W	R	I	S	U	R	H	X	U	D	E
H	D	F	D	G	P	H	R	H	P	G	R	G	P	F	H	Q	P	A	L	S	Q	A	O	Q	Q	G	R	Q	C	R
G	P	E	C	F	Y	G	N	G	Y	F	N	F	Y	E	L	N	Y	S	J	Z	I	S	U	N	I	C	N	N	O	D
F	X	Z	B	E	R	P	M	F	R	B	M	E	R	Z	C	C	R	O	I	Q	L	O	T	M	L	F	M	C	B	C
E	K	R	A	D	K	F	L	E	K	E	L	D	K	R	B	M	K	T	E	V	K	T	P	L	K	T	L	M	K	B
D	C	D	S	C	Z	E	J	D	Z	P	J	C	V	D	A	L	Z	P	H	O	Z	P	Y	A	Z	P	J	L	Z	V
C	B	P	O	Z	X	D	I	C	X	Y	I	Z	Z	P	S	J	X	Y	S	T	X	Y	J	J	X	Y	I	J	X	X
B	A	X	T	B	W	X	H	B	W	R	H	B	L	X	O	T	W	R	O	P	W	R	M	I	W	R	H	T	W	A
A	S	K	P	A	V	K	G	A	G	K	G	A	J	K	T	S	V	K	T	Y	N	K	R	H	N	K	G	S	V	K
67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97
V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R
S	D	C	L	S	P	C	W	W	P	Z	W	S	P	C	E	I	P	Z	V	X	Q	Z	V	Y	Q	Z	W	I	C	U
V	P	B	F	Q	Y	Y	S	S	Y	X	S	Q	Y	B	G	B	Y	I	U	R	I	X	H	T	I	X	S	B	O	W
O	X	A	H	V	R	W	Z	O	R	W	Z	V	R	A	V	P	R	X	Q	N	L	H	G	G	L	W	Z	P	B	S
T	K	S	W	O	K	Z	Q	T	K	V	Q	O	K	S	I	O	K	W	B	M	K	W	F	S	K	V	Q	O	K	O
P	C	Y	V	X	Z	B	V	M	Z	U	V	X	V	Y	D	H	Z	V	D	L	Z	V	E	R	Z	U	V	H	Z	T
L	B	O	U	N	X	A	O	Q	X	Q	O	N	Z	O	F	G	X	D	N	J	X	U	D	P	X	E	O	G	X	P
N	A	W	Q	U	W	S	T	P	W	D	T	U	L	W	U	F	W	F	F	I	W	C	C	F	W	Q	T	F	W	Z
Y	S	V	J	T	V	O	P	V	G	N	P	T	J	V	Q	E	V	U	A	H	N	E	B	E	N	N	P	E	V	Y
U	Y	U	N	P	F	R	Y	L	V	M	Y	P	I	U	N	A	U	H	C	G	V	Q	A	D	V	M	Y	A	U	Q
J	O	Q	E	Y	U	T	X	N	U	L	X	Y	H	Q	M	D	Q	C	M	F	H	G	S	C	H	S	X	D	Q	N
M	W	N	D	R	Q	V	R	Y	Q	O	R	R	G	N	H	Y	D	E	L	E	G	B	O	O	G	B	R	Y	N	M
R	V	M	C	W	N	U	N	R	B	A	N	W	F	M	L	R	N	Q	J	D	F	D	U	B	F	L	N	R	M	L
K	U	L	B	K	A	Q	M	K	D	J	M	K	E	L	C	K	M	N	I	K	E	N	T	K	E	D	M	K	L	J
Z	Q	J	A	M	C	N	L	Z	N	C	L	M	D	J	B	Z	L	M	E	C	D	M	P	Z	D	O	L	Z	A	I
X	N	I	S	L	M	M	J	U	F	T	J	L	C	I	A	X	O	L	H	B	C	L	Y	X	C	A	J	X	J	H
Q	M	H	O	J	E	L	I	X	A	S	I	J	B	H	S	W	A	G	S	A	B	J	J	W	B	J	I	W	I	G
W	L	T	T	I	S	J	H	J	C	I	H	I	X	T	O	V	J	J	O	U	A	F	M	V	A	I	H	V	H	F
I	J	G	P	H	B	I	G	I	M	H	G	H	A	G	T	U	C	B	T	W	S	I	R	U	S	H	G	U	Y	E
H	I	F	Y	G	L	H	F	H	L	G	F	G	S	F	P	Q	T	A	P	S	O	A	Q	Q	O	G	F	Q	T	R
G	H	E	M	F	J	G	E	G	J	F	E	F	O	E	Y	N	S	S	Y	Z	U	S	I	N	U	C	E	N	G	D
F	T	Z	R	E	I	P	D	F	I	B	D	E	N	Z	R	C	I	O	R	Q	T	O	L	M	T	F	D	C	S	C
E	G	R	K	D	H	F	K	E	E	K	D	U	R	K	M	H	T	K	V	P	T	K	L	P	T	K	M	R	B	
D	F	D	Z	C	D	E	C	D	H	P	C	C	T	D	J	L	G	P	Z	O	Y	P	Z	A	Y	P	C	L	P	V
C	E	P	G	Z	G	D	B	C	S	Y	B	Z	W	P	Z	J	F	Y	X	T	J	Y	X	J	J	Y	B	J	F	X
B	Z	X	I	B	O	X	A	B	O	R	A	B	M	X	X	T	B	R	W	P	M	R	W	I	M	R	A	T	E	A
A	R	K	X	A	T	K	U	A	T	K	U	A	Q	K	W	S	E	K	G	Y	R	K	N	H	R	K	U	S	D	K

DIID pattern #13 AUTOGE (50% of AUTOGENERATE?), #26 NORTHE (NORTH E! in the right place!), #31 HOTEL, #63 SIGN, #64 BER + #67 MIN (83% of BERLIN in the right place. Also the L is shift -1 of M in the alphabets, and we have found LIN in the right place in the IID pattern) and #83 KIND.

Tests:

1 Using an algorithm, derived from the pattern of K4 sequence DIAWINFBN, which uses both alphabets, and expanding the same with the interpretation of I and D BY ROWS; we got the correct plaintext of BERLIN and NORTH E, and the correct key (ELYOIE) of BERLIN. This tests that IDBYROWS is a real clue and that the TFJ algorithm would be very useful for finding clues of words that are part of the K4 plaintext. I have not yet built tables with IK, DK, RQ or SOS patterns, maybe there could be new clues in them. The TFJ algorithm to become the Kryptos algorithm, we must either modify it in some way or apply it in a different way.

If we include the 2 EAST, we have ZAQI NORTH EVBY, a correct character in each EAST, that is, only 54% of the correct plaintext. How should the operative in the field do to get the correct plaintext? One possibility would be, that he deduces NORTH EAST from the clue NORTH E. And then he searched all of Kryptos for references to NE. At the coordinates of K2 there are N and W. At the compass, the needle points to the lodestone, WSW. But the other end of the needle points to ENE, EAST NORTH EAST. That possibility does not seem very good but ... maybe Kryptos will give us the instructions or the clues as before.

2 The 1-to-1 relationship between ciphertext and plaintext begins to be confirmed, just as Jim told us.

3 Clues of the transpositional and substitution code classes mentioned in Pre-Kryptos appear. TRANSPSWED may be referring to the K4 transposition that we saw in the M1 message (LINE BLUE, tested 4 times) and AUTOGE may be referring to these substitution algorithms that we are working with, where the key is auto-generated from positions, jumps and the ciphertext itself.

Some modulo 26 arithmetic to better understand the algorithm. It's just additions and subtractions, no rocket science here. The Vigenere or Quagmire III ciphers (like the one used in K1 and K2), if we use the ordinal numbers of the alphabet, can be expressed like this:

Cipher $P(i) + K(i) = C(i)$ Decipher $C(i) - K(i) = P(i)$ Using the Kryptos alphabet

Very simply, for each character of the plaintext P that occupies place i, the key K corresponding to place i is added to it and the ciphertext character C of place i is obtained. Something similar happens when we decipher.

For the algorithms we are using it is a little longer to explain. I'm going to use Tip T instead of Pointer so it doesn't get confused with plaintext P.

Decipher $T(i-1) - J(i) + C(i) = K(i)$ Using the ABC alphabet

It is also simple, the pointer or Tip of the previous place T(i-1) is taken, the jump J(i) corresponding to the place i is subtracted, the cipher character C(i) of the place i is added and we obtain the key K(i) of place i. Always using the ABC alphabet. Then, using the Kryptos alphabet, with the key K(i) and the cipher C(i) we obtain the plain P(i).

These are asymmetric algorithms, in the sense that we can always decipher but we cannot always cipher. To decipher given a pointer or tip T, a jump J and a code C we can always find a key K and therefore a plain P. But to cipher it is not so simple. We must find a pair K(i) and C(i) that with the alphabet ABC complies with:

$$C(i) - K(i) = J(i) - T(i-1)$$

And at the same time with the Kryptos alphabet complies with:

$$C(i) - K(i) = P(i)$$

Depending on the P(i) that we want to cipher and the corresponding jump, there are times when there is no K(i) and C(i) pair that meets both conditions. That is why I have the theory that this algorithm only gives us clues to find parts of the K4 plaintext or we must modify it in some way or apply it in a different way. Unless the plaintext has been chosen so carefully that K and C pairs that satisfy the conditions always correspond.

With Anders' recent observation that part of the words appear correct and the parts that are not correct have some shift, I checked the sequence #32 SOS PALIMYSDTO and indeed the wrong characters are very close to the correct ones in the Kryptos alphabet: Y+1 = P, D+1 = E, T+2 = S and O-1 = T. So #35 PALIMPSEST is a good candidate for plaintext and its place is next to EASTNORTHEAST. It is as if there is a small noise at the end of the words. Is this the way Jim found to modify the algorithm and hide the true plain text from Ed? Or will it be the confirmation that when there is no K and C pair that meets the two conditions, it changes +-1 or +-2 places in the Kryptos alphabet so that the operator in the field can still recognize the plaintext? These two alternatives still do not convince me, but I am confident that Kryptos, as it has done so far, will tell us how to do it.

Finally, we see two new examples of duress ciphers. In the IDID pattern place #26, as we mentioned, TRANSPSWED appears, but that is the place where Jim confirms NORTHEAST. In the DIID pattern place #31 HOTEL, but that is the place where Jim confirms the second EAST. Are there two plaintexts interlaced? With the same ciphertext, changing the jump pattern or the initial pointer, we get two different plaintexts, both valid, both useful. In the M7 message I will show how NORTH HOTEL leads us to a new doorway and to one of the logical explanations of the DYHAR cluster.

This has been the fifth message of ten related with K4 and K5, and there is still much more!

Jose Luis